

Biotech Highlight

Soft sensors in bioprocessing: A status report and recommendations

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The following report with recommendations is the result of an expert panel meeting on soft sensor applications in bioprocess engineering that was organized by the *Measurement, Monitoring, Modelling and Control* (M³C) Working Group of the European Federation of Biotechnology – Section of Biochemical Engineering Science (ESBES). The aim of the panel was to provide an update on the present status of the subject and to identify critical needs and issues for the furthering of the successful development of soft sensor methods in bioprocess engineering research and for industrial applications, in particular with focus on biopharmaceutical applications. It concludes with a set of recommendations, which highlight current prospects for the extended use of soft sensors and those areas requiring development.

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1 Background of the workshop

Measurement, monitoring, modelling and control (M³C) are critical to biochemical engineering due to the complexity inherent in the biological systems and molecules used [1–3]. The recent interest in

M³C methodology from the regulatory authorities for pharmaceuticals has been clearly stated in the Food and Drug Administration (FDA) and European Medicines Agency (EMA) guidance and recommendation documents [4–5], as well as in the quality guidelines from the International Conference on Harmonisation (ICH) [6].

Although these documents are directed towards the pharmaceutical and biopharmaceutical manufacturers, it is also stated clearly that the academic community has important roles as inventors and developers, and as educators promoting the general acceptance of new technology developments [7–8].

The expert panel focused on mapping how the academic and industrial competences in M³C can concertedly contribute to further development of better soft sensor solutions for purposes outlined in guidance documents and in a broader sense by

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Abbreviations: ANN, artificial neural network; EMA, European Medicines Agency; FDA, Food and Drug Administration; HPLC, high-performance liquid chromatography; ICH, International Conference on Harmonisation; M³C, Measurement, monitoring, modelling and control; MIR, mid-infrared; MVDA, multivariate data analysis; NIR, near-infrared; PAT, process analytical technology; PCA, principal component analysis; PLS, partial least square; QbD, quality by design

considering other biotechnology manufacturing situations. Thus, although the focus is on pharmaceutical manufacturing, other fields of bioprocessing, e.g. of industrial enzyme manufacture, food processing, and solid state fermentations, may also benefit from the conclusions and recommendations provided. This expert report provides an overview of the present situation and makes recommendations and suggestions to address current and future needs. The attending experts were selected due to their experience in and the knowledge of the area's scientific and industrial requirements.

2 What is a soft sensor? A definition

The term *soft sensor* or *software sensor* has appeared repeatedly in industrial process monitoring [9–17]. The term combines the words “software”, because the sensor signal evaluation models are usually implemented in computer programs, and “sensors”, because these models are delivering information similar to hardware sensors [9]. The principle of a soft sensor is illustrated in Figs. 1 and 2. The figures imply that the soft sensor is capable of on-line monitoring. On-line may be interpreted as measurements provided by probes, sensors, devices or instruments in or in the vicinity of a bioreactor or any other bioprocess unit operation where the signals are capable of just-in-time monitoring.

An example of a soft sensor application that adheres to this definition was recently suggested by Warth et al. [17]. They apply a soft sensor set up very similar to the configuration in Fig. 2B. A recombinant *Escherichia coli* fed-batch culture producing green fluorescent protein is monitored by an on-line near-infrared (NIR) probe placed in situ in the bioreactor that assesses the biomass concentration from the absorption at 1100 nm, and a filtration probe that withdraws media samples from the bioreactor for on-line analysis of glucose and acetate concentrations in an HPLC. Soft sensor algorithms compute the specific growth rate of the biomass from the NIR signal and the specific glu-

cose uptake rate and acetate formation rate from the NIR and HPLC data. The data are continuously updated and presented to the process operator on the control screen.

The model of the soft sensor is often of a predictive character. Using the term *inferential sensors* [18], *virtual on-line analyzer*, (as referred to in the Six-Sigma context [19]) or *observer-based sensors* [20] may also seem appropriate.

The central idea behind a soft sensor is to use (relatively) easily accessible on-line data for the estimation of other cultivation variables that are otherwise either difficult to measure or only measured at low frequency. Alternatively, soft sensors can also be used within process monitoring, e.g. with the purpose of performing fault detection and diagnosis. At a very general level one can distinguish two different classes of *soft sensors*, namely model-driven and data-driven [9]. Chemometric techniques such as partial least square (PLS) belong to the second class of soft sensors, and can also be applied to traditional process data [16]. Of course, there are other data-driven soft sensors, and an exhaustive overview of methods frequently employed in process industries can be found in [9]. Specifically for application of data-driven soft sensors in pharmaceutical production processes, it is important to note that soft sensors based on PLS or principal component analysis (PCA) are generally preferred, since these methods are well-known in the pharmaceutical industry and this facilitates validation. Model-driven soft sensors can be based on mass or energy balances (first principles approach). The Kalman filter and the extended Kalman filter methods belong to this class. The main drawback is that the development of such a model-driven soft sensor is generally considered to be rather time consuming [9]. When developing a pharmaceutical production process, time is usually one of the most critical factors, and that in our view is one of the main reasons why such model-driven soft sensors are not more widespread in the pharmaceutical industry.

Significant interest in soft sensors appears in several engineering areas and associated industri-

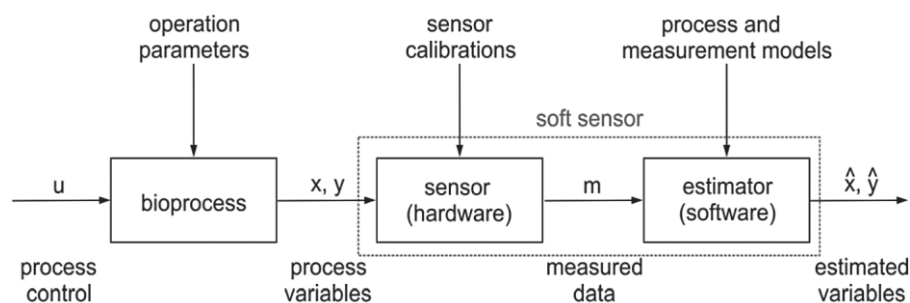


Figure 1. The soft sensor principle as defined in [10, 11]. The authors frequently use the term *software sensor* as well for this principle. The figure shows one hardware sensor and one estimator. In reality these can be a combination of several hardware sensors.

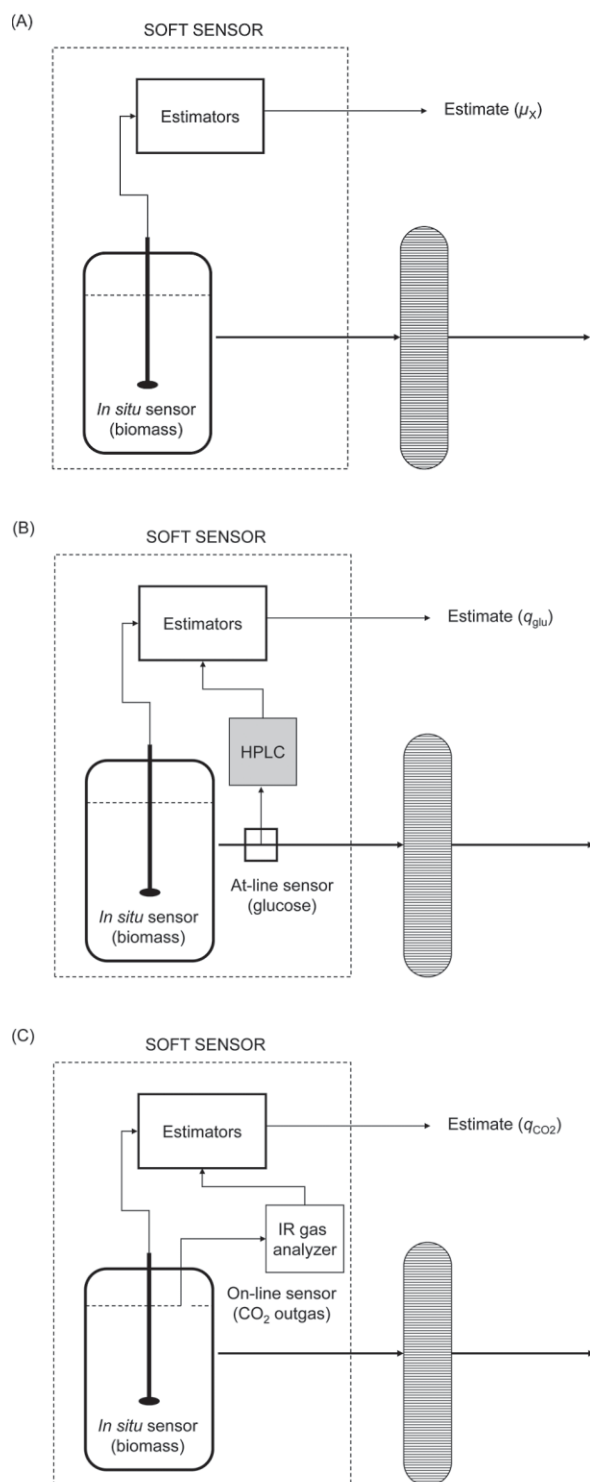


Figure 2. Schematic examples of soft sensor configurations in a bioprocess. (A) A soft sensor where specific growth rate in the bioreactor is estimated by an in situ sensor for biomass concentration. (B) A soft sensor where the specific glucose uptake rate is estimated using an in situ biomass sensor and an at-line HPLC for glucose concentration. (C) A soft sensor where the in situ sensor is combined with an off-gas infrared measurement for exit carbon dioxide determination for estimating specific carbon production rate.

al applications such as lyophilisation, distillation, wastewater treatment [11, 12], and, of particular interest in this report, biotechnology [13–17].

3 Scientific and technological state-of-the-art and challenges

The state-of-the-art of this class of analytical device is directly dependent on the availability and capability of the two main components, i.e. the measurement device(s) and the mathematical process model(s), which, when successfully merged into one functional entity, comprise the soft sensor. Moreover, the state-of-the-art of these components sets the limit of the entire soft sensor configuration.

It is therefore interesting to detail the potential options that measurement technology and mathematical modelling provide for the development of soft sensor technology. This will be contrasted with the currently reported soft sensors in order to illuminate the extent to which this potential has actually been exploited. Clearly the challenge is to develop the unexplored possibilities.

3.1 Measurement technology

The literature reveals that a wide variety of measurement techniques have been applied to monitoring bioprocesses [1–3]. Table 1 summarizes prominent and frequently reported results. The table categorizes the measurement techniques from a perspective that highlights their utility for on-line monitoring, thereby paving the way for merging with modelling methods to create soft sensors.

On-line measurement techniques are based on in situ procedures which are divided in in-line sensors, such as invasive probes (e.g. temperature, pressure, turbidity probes) that are placed directly in the media of a bioreactor or a process stream of any other process operation and on-line devices which are used in close proximity of the bioprocess medium or stream (e.g. mass flow controller, off-gas analyzers, spectrometers) [4].

Several of these measurements may play an important role in carbon and oxygen balances in the cultivation process [21]. Volatile compounds, such as methanol and ammonia in off-gas may be related to liquid phase concentrations, thereby supporting the balances.

These approaches are sometimes combined with non-invasive on-line spectroscopic methods, where spectral information is acquired from 2D fluorescence, NIR/MIR (mid-infrared) and Raman spectral data which can be used for process moni-

Table 1. State-of-the-art on-line measurement techniques for soft sensors

Measurement device	Examples of analytes or state variables
In situ sensors/analyzers:	
Physical in situ sensors/probes	Temperature, pressure, volumetric or mass flow rates (gas & liquids), weights
Electrodes:	
NIR/MIR in situ probes	Media components, biomass
Multiwavelength fluorimetry	NAD(P)H, amino acids
Turbidimetry	Biomass
In situ microscopy	Cell shape etc.
radio-frequency-impedance	Biomass, cell viability
At-line sensors/analyzers:	
Liquid phase analysis:	
High performance liquid chromatography	Media components (organic acids, amino acids, saccharides), recombinant proteins
Flow injection analysis (FIA) with various detection methods	Media components, extracellular products
Electrode-based biosensor	Enzymatic substrates (e.g. media components)
Immuno-based biosensor	Antigens (e.g. proteins)
Flow diffusion analyzer	Glucose, methanol
In-line microscopy	Image analysis
Coulter counter	Cell number, cell size distribution
Flow cytometry	Cell viability, cell constituents (e.g. DNA, RNA, proteins, lipids, membrane characteristics, intracellular pH, redox)
Off-gas analysis:	
Paramagnetic oxygen analyzers	O ₂
Infrared adsorption photometers	CO ₂
Gas chromatographs	O ₂ , volatile compounds
Mass spectrometers	Low molecular weight compounds
Flame ionization detectors	CH ₃ OH, CH ₄
Electronic noses	Microbial volatiles (ethanol, sulfides), infectants

toring and modelling with multivariate data analysis (MVDA) [22–24].

At-line measurement may exploit analytical chemistry methods for bioprocess stream analysis either in closed or open bypass flow procedures [25, 26]. In particular HPLC can be very useful for the analysis of target molecules, side-products and media components. Great success has been achieved with recombinant proteins, (e.g. regulatory proteins, vaccines, enzymes) where purification methods of proteins are frequently based on HPLC-procedures [27, 28].

The number of analytes and measurements available in-line, on-line, or at-line increases rapidly and opens up a possibility of using these in soft sensors. For the soft sensor concept, the in situ sensors may seem more attractive than at-line sensors. However, precision, calibration and stability as well as the general analytical characteristics often favour the at-line alternatives.

3.2 Mathematical modelling of measurement data

Mathematical modelling of bioprocess data has been discussed in biotechnology for the past 30 years [21]. These models may generally be categorized into steady-state and dynamic models.

Steady-state models are frequently developed from mass and component balances, from mass or heat transfer laws or even from elemental balances. Purely empirical black-box-models with constant parameters usually also belong to the group of steady-state models.

Dynamic models usually consist of dynamic balances in combination with kinetics to describe rate expressions as functions of the state variables. These models may be classified as unstructured and structured models. The major drawback of unstructured models in the context of soft sensors is that they use constant yield coefficients. Structured models, on the other hand, can be used to account

for changing yield coefficients and to capture more complex behavior in cultivations. Although structured models can describe a wide range of very complex cultivation behaviour, unstructured models are more frequently used within estimator- or observer-based soft sensors, due to their relative simplicity. Structured models, however, are applied for the calculation of optimal feeding profiles and other control strategies [29].

Models with a high potential for use in soft sensors are often based on artificial neural network (ANN) and fuzzy sets [30]. ANNs are very powerful due to their high flexibility. They may be used as multi-input single-output or even multi-input/multi-output models. Fuzzy sets are based on general rules and they have also been shown to be capable of describing unknown state variables from known measurements.

3.3 Existing applications

Existing soft sensor applications based on the above mentioned devices and models are, so far, relatively few. Table 2 lists some of these. The soft sensors are mostly based on on-line measurements from off-gas analyzers [31]. Quite common is the simple calculation of derived variables such as the oxygen uptake rate, the carbon dioxide evolution

rate, the respiratory quotient and the oxygen transfer coefficient ($k_L a$). More sophisticated techniques using estimators or observers are used to estimate biomass concentrations, growth rates, substrate and product concentrations or substrate uptake and product formation rates. If the stoichiometry of the general process reaction is changing during the cultivation, additional information from at-line analyzers; titrant volume etc. has been used to improve the modelling accuracy.

The development of new and more robust on-line analytical techniques will lead to improved soft sensors, delivering more precise estimates of important parameters, such as the growth rate and product formation rates, even when off-gas analysis may be difficult. One route to achieving this may be the use of sensor arrays in combination with pattern recognition systems in the gas phase (electronic nose) [32] and the liquid phase (electronic tongue) [33] for monitoring and control of bioreactors.

The requirements for soft sensors in downstream processing focus on the product and critical contaminants [34]. The most desired measurement is that of the product molecule, however, in the production of biologics this first requires a clear definition of what this means as scope for considerable product heterogeneity exists. Some product forms

Table 2. Soft sensor applications

Soft sensor application	Measurement devices	Models	Ref
μ -estimation for control	Gas analyzers for O_2 and CO_2	Mass balance	[25, 41]
μ -estimation with base titration	pH electrode, titrand volume	Mass balance	[15]
μ - and q_p estimation	NIR, at-line HPLC	Mass balance	[20]
Estimators/Observers			
Estimation by indirect measurements and correlations: OUR, CER, RQ, $k_L a$ -estimation	Gas analyzers for O_2 , CO_2 , dissolved O_2 -probe	Mass balances, mass-transfer equation	[22]
Estimation through macroscopic balances: stoichiometric coefficient estimation	Gas analyzers for O_2 , CO_2 , titration, substrate conc., biomass composition (offline)	Mass balances, elemental balances, unstructured models	[22]
Extended Kalman filter for biomass, substrate and product concentrations, yield and growth rate estimation	Gas analyzers for O_2 and CO_2 as well as measurement for nitrogen uptake rate	Dynamic mass balances, unstructured models	[22]
Asymptotic observer to estimate C-source, N, Biomass, product	Dissolved O_2 -probe, flow $O_{2,in}$, flow $O_{2,out}$, volume, inlet fructose and nitrogen concentration (offline)	Dynamic component balances	[22]
Online detection of cell mass, viable cells, glucose, ammonia, acetate, and cell internal product	Gas analyzer for O_2 and CO_2 , radio frequency impedance, 2D-fluorescence spectroscopy, FIA, HPLC	Dynamic mass balances, PCA, ICA, MISO, MIMO, ANN	[35, 36]

ICA, independent component analysis; MIMO, multi-input/multi-output models; MISO, multi-input single-output models

will be considered product related impurities (i.e. multimers or fragments of the product or undesired chemical modifications of the product such as differently glycosylated or deamidated and oxidized forms). This makes direct measurement via a soft sensor challenging; success has been reported using biosensor [35] and Fourier transform infrared spectroscopy approaches [36]. Such methods usually only provide relatively coarse grain understanding of the product, while more detailed aspects of the product structure would be measured off-line in a quality control laboratory often using mass spectrometry. The area is under investigation and Raman spectroscopy-based methods have been shown to measure glycosylated states in simple systems [37].

3.4 Transfer of modeled data

In order for a soft sensor to be applied to an industrial process, data transfer and handling of data must be addressed.

For set up and parametrization, a soft sensor relies on certain process data, e.g. trial runs, lab analyses or modelling data. These are usually historical data, which implies the need of some batch-based process data archive. Mandatory in regulated environments, current industrial automation systems provide such archives, though the sheer amount of data is a limiting factor for large-scale processes, both in storage and communication capacity.

A soft sensor may be realized within the automation system of the process. Yet the computational capability of automation hardware is quite limited, and the automation systems focus predominantly on safe, reliable and robust process control rather than on numerical or statistical computations. In short, soft sensors based on more than a few basic calculations should be separated from the automation system and run on some dedicated computing hardware. Communication will be realized via Industrial Ethernet or some field bus. Communication should be limited to information rather than data, i.e. the soft sensor should provide the particular result of soft sensing such as the single value of protein concentration. Of course, these results should again be stored batch-wise in the process data archive.

4 Industrial needs and opportunities for soft sensors in biochemical engineering

The complexity of the bioreaction and downstream steps involved in bioprocessing and the need to fa-

cilitate process monitoring of these steps have propelled a strong interest in the soft sensor technology within this industrial sector for years [13]. Early work on soft sensors in the bioprocess industry concentrated on on-line prediction of time trajectories of process variables that were traditionally difficult or time consuming to measure, such as biomass or product concentrations [10]. These sensors were typically using on-line/in situ measurements of exit gas composition and physico-chemical parameters, such as pH, temperature and dissolved oxygen [38]. They were frequently applied in various bioprocesses, ranging from microbial to cell culture cultivations, for monitoring, on-line process control and fault detection with the aim of improving productivity and process efficiency [39–40]. These applications were predominantly implemented using dedicated computing hardware communicating with data warehousing systems used by individual companies, which is in line with the observations described in the previous section.

Following the wider industrial acceptance of spectral analytical techniques, such as NIR spectroscopy or fluorescence measurements, applications of soft sensors expanded to on-line estimation of multiple metabolites [16].

Quality by design (QbD) and process analytical technology (PAT) initiatives present a new impetus for the application of soft sensors, in particular in the highly regulated biopharmaceutical industry. The complexity of the process dynamics, the variability resulting from raw materials and predominantly batch processing as well as the ever increasing pressures of reducing lead times of new products to market increase the interest in soft sensor approaches both in process development and on-line monitoring and control of production. There are also strong needs evolving in industrial biotechnology, a sector driven by the need to find sustainable process alternatives to oil- and gas-based processes. This will likely require the productivity improvements brought by continuous processing to be economically viable. This brings about an increased need for robust measurement, monitoring, modelling and control.

There are a number of important requirements placed upon a soft sensor in such a challenging application area (see also list in Table 3). Amongst these requirements, minimum re-calibration required within batch, between individual batches or cultivation vessels is very important as it increases the productive time of the soft sensor during processing. Soft sensors based on, for example, NIR have been shown to require the application of various data pre-processing techniques to increase their applicability [41]. Additional challenges aris-

Table 3. Critical needs in industry and consequences for soft sensors

Need	Consequences of the need for soft sensor implementation
<i>Operational performance of the soft sensor</i>	
Long-term stability	Must allow recalibration during operation or exhibit high repeatability
Short-term response	Sample transfer and detector response have short time lags
Easy recalibration	Done once, automatic or not necessary
Highly reliable operation	Should not require constant operator actions, recalibration
Highly automated	Few mechanical parts
	Optical and electronic systems favoured
Multi-analyte capacity	Spectroscopy methods (NIR/MIR) and chromatographic methods favoured
<i>Process economics related</i>	
Monitor productivity	Main product detected
Process efficiency	
Low maintenance cost	Very robust hardware sensor
Moderate capital investment for sensor	Too complex hardware avoided. Still software can be complex beneath the user interface.
Low operational cost	High degree of automation. Few reagents if any. Few consumables.
<i>Specific needs</i>	
Monitor seed quality	
Monitor raw materials	
<i>Regulatory needs</i>	
Monitor variability	Analytes defined
Monitor deviations	Just-in-time
Monitor CQA (QbD)	
Monitoring downstream impurities	

ing from the nature of the analytical measurements used in these soft sensors include drifts and outlier/missing data handling procedures that are frequently required to achieve acceptable performance of the soft sensors in an industrial environment.

Provided such issues of operability are addressed satisfactorily, soft sensors will be industrially highly beneficial in a range of applications. Assessment of raw material quality is particularly important for processes where complex raw materials with inherent variability can lead to batch to batch variability affecting the overall process efficiency. Examples of successful applications of soft sensors in raw material assessment in industrial manufacturing bioprocesses have been reported in a range of bioprocesses [42–43]. The quality of seed cultivations used to inoculate production vessels is equally important in any bioprocess involving staged production or sub-passaging of culture [44]. The industrial applications of soft sensors in on-line monitoring and deviation detection have already been highlighted above with relevant examples of literature sources. However, it is also important to identify the applicability of soft sensors in downstream process optimisation [45] and in particular the need for overall process optimisation based on

soft sensors as well as mechanistic models of individual unit operations [46]. In such optimisation frameworks, the performance of individual soft sensors and unit operation models is assessed in light of their contribution to the overall process performance estimation with the view of achieving optimum process output, rather than selecting the model/soft sensor providing the best performance on individual unit operations in isolation. Given the widely acknowledged influence of unit operations upon subsequent processing and the potential of variability upstream significantly impacting the product quality and quantity, such a framework would provide industry with a much needed tool for building quality into the final product.

5 Conclusions and Recommendations

In conclusion, the M3C makes the following recommendations:

- i. It is recommended that the term *soft sensor* is used for on-line monitoring applications where hardware measurement devices, such as invasive in-line probes, non-invasive spectrometric analyzers, or at-line flow injection methods generate signals that are directly

processed in appropriate computational models or algorithms that are implemented in software solutions. The hardware part of the soft sensor may be single small sensor devices or more elaborate analytical setups where pumps and separation columns are integrated. The hardware device may be a single unit or a combination of several devices. The response time may be very short (real-time) or delayed, due to operational procedures in the setup.

- ii. Soft sensors are widely used in other industrial applications and not only bioprocesses. Examples are automotive and aircraft control, wastewater treatment plants, industrial scale distillation. Soft sensors for bioprocesses can adapt measurement technology from these other industrial branches. A potential challenge with introducing new types of soft sensors in industry is the acceptance by the regulatory authorities (e.g. FDA, EMA). This is exacerbated even further with more sophisticated soft sensors, such as for example adaptive soft sensors, where the performance of the soft sensor is monitored and adapted in real-time to improve its performance. Regulatory validation of such adaptation during process operation is currently perceived as particularly challenging.
- iii. Soft sensors should aim at a simplification of production. The soft sensor should not result in a more complex procedure at the manufacturing site (e.g. the number of SOPs should not increase but become fewer, and the need for training of operators should not increase significantly).
- iv. Knowledge and information transfer about soft sensors from research, academic as well as industrial R&D, to manufacturing units should be enhanced with the objective of increasing awareness of the potential of soft sensors. This concerns both the hardware sensor alternatives as well as the software methodologies available.
- v. Implementation methods for application of soft sensors should be standardized and be as simple as possible. Simulation techniques should support soft sensor development, where in silico soft sensor development and testing on a process model can reduce the downtime of the production due to a more efficient soft sensor implementation on the bioprocess. Process data should be archived batch-wise to support modelling, setup, and parameterization of soft sensors.
- vi. Soft sensors with low computational and communication requirements should be seamless-

ly fitted into industrial automation systems. Soft sensors with higher demand on calculations and data transfer should be implemented independent of process automation and provide their results by standard protocols such as Industrial Ethernet or fieldbus.

- vii. New soft sensors should be developed within a specific context. For example, the industrial environment should be clearly defined, or the production strategy should be clearly stated.
- viii. A number of case studies based on industrial data should be carried out and documented with the purpose of benchmarking soft sensors. These cases should represent typical bioprocesses of current interest (e.g. mammalian cell production processes, secondary metabolite production, industrial enzyme production, and vaccines).

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6 References

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